

DATA ANALYSIS PROJECT REPORT

Topics: Pandas, NumPy, and Matplotlib

Project Title	Analysis of the Rental Housing Market in Dhaka
Student Name	Sabbir Hossain Niyaz
Student ID	22-47538-2
Course / Section	Programming in Python
Instructor	Md. Tanzeem Rahat
Date of Submission	3 May 2025
Dataset Source	Kaggle — House Rent of Dhaka and Chittagong (bproperty)

1. Executive Summary

This project investigates the residential rental housing market in Dhaka, Bangladesh, using a real-world property listing dataset scraped from bproperty.com and published on Kaggle. The primary goal was to answer three evidence-based analytical questions: which areas in Dhaka command the highest average rents, how strongly property size (square footage) predicts rent, and how bedroom count affects price and value. The dataset was filtered to Dhaka-only listings and underwent five data-cleaning steps, including missing-value removal, duplicate elimination, and type conversion. Three engineered features, price per sqft, rent category, and bedroom group. Were created to support deeper analysis. Key findings reveal that premium neighbourhoods such as Gulshan, Banani, and Baridhara sit significantly above the city-wide median rent, that a positive but imperfect correlation exists between property size and rent, and that higher-priced listings charge more per square foot, not merely more in absolute terms. NumPy was used for z-score computation, normalisation, percentile analysis, and trend-line fitting. All three research questions were answered with visual and statistical evidence.

2. Introduction and Project Objectives

Dhaka is one of the most densely populated cities in Asia and faces a chronic housing affordability crisis. Rental prices vary sharply across neighbourhoods, and prospective tenants, policymakers, and property investors often lack accessible, data-driven insight into how prices are distributed and what factors drive

them. This project uses a real dataset of property listings to provide evidence-based answers to questions that matter to ordinary renters in Dhaka.

Item	Detail
Dataset Name	House Rent of Dhaka and Chittagong (bproperty)
Dataset Source	https://www.kaggle.com/datasets/shoaibshahriar33/house-rent-of-dhaka-and-chittagong-bproperty
Main Project Goal	To analyse the Dhaka residential rental market using Python, identify the key drivers of rent, and answer three specific analytical questions using Pandas, NumPy, and Matplotlib.
Research Question 1	Which areas in Dhaka have the highest average rental prices, and how large is the price gap between expensive and affordable areas?
Research Question 2	How strongly does property size (square footage) influence rent, and is there a clear linear relationship between the two?
Research Question 3	How does the number of bedrooms affect rent, and which bedroom category offers the best value when measured by price per square foot?

3. Dataset Description

The dataset contains real residential property listings scraped from bproperty.com, one of Bangladesh's largest online property marketplaces. Each row represents a single rental listing. Available columns include the area name, number of bedrooms (beds), property size in square feet (sqft), city/location, and monthly rent in Bangladeshi Taka (BDT). For this project, the dataset was filtered to include only Dhaka listings, as Dhaka's rental market has unique characteristics driven by its status as the national capital and commercial centre.

3.1 Basic Dataset Information

Property	Value
Number of rows (post-filter)	Dhaka-only subset (original dataset: 2,000+ listings)
Number of columns	5 core columns: Area, Beds, Sqft, Location, Rent
Data types present	Numerical (rent, beds), String/Object (sqft — required cleaning), Categorical (Area, Location)
Target / key variable(s)	rent (monthly rental price in BDT)

Issues observed at first glance	sqft stored as string with non-numeric characters; missing values in multiple columns; beds column stored as object type; potential duplicate listings from re-posted ads
--	---

3.2 Initial Observations

On first inspection using `df.info()` and `df.head()`, the most critical issue was that the `sqft` column was stored as an object (string) rather than a numeric type, because the raw values contained characters such as commas and the text 'sqft'. This prevented any mathematical computation on property size. Missing values were present in several columns, and the `rent` column showed a very wide range, suggesting extreme outliers at both ends of the market. The `beds` column also required type conversion before it could be used in numerical analysis.

4. Data Cleaning and Preparation

The following cleaning steps were performed before any analysis. Each step is justified by the problem it solved and its impact on the integrity of subsequent results.

Issue Found	Action Taken	Reason / Justification	Code Reference
Missing values across key columns (area, beds, rent, sqft)	Dropped all rows with any missing value using <code>dropna()</code>	Rows missing rent, sqft, or beds cannot contribute to price analysis. Imputing rental prices would introduce false data and mislead comparisons.	Step 4 — Cell 1
Duplicate listings from re-posted property ads	Removed all exact duplicate rows using <code>drop_duplicates()</code>	Duplicate entries artificially inflate listing counts and skew average rent calculations. Each property should be counted once.	Step 4 — Cell 1
sqft column stored as string with	Applied regex substitution to strip	Calculating price per sqft and performing correlation	Step 4 — Cell 1

non-numeric characters (commas, 'sqft' text)	non-numeric chars, then converted to float using <code>pd.to_numeric()</code>	analysis both require sqft to be a numeric type. Without this step, all size-based analysis fails.	
The rent column may contain non-numeric placeholders	Explicitly converted rent to numeric using <code>pd.to_numeric(errors='coerce')</code> and dropped resulting NaN rows	A safety step to ensure the primary analysis variable is clean. Any text placeholders become NaN and are safely removed.	Step 4 — Cell 1
Properties with zero or physically impossible sqft values	Filtered out rows where <code>sqft <= 0</code>	A property cannot have zero square footage. Such entries are data entry errors. Keeping them causes division-by-zero errors in the <code>price_per_sqft</code> calculation.	Step 4 — Cell 1

Outlier removal (Step 5): The IQR method was applied to remove statistically extreme rent values. Ultra-luxury listings priced more than $1.5 \times \text{IQR}$ above Q3 were excluded to prevent them from distorting area-level averages. This step is acknowledged as a limitation — see Section 10.

5. Feature Engineering

Three new features were created from existing columns to support the analysis and answer the research questions more effectively.

New Feature	How It Was Created	Why It Is Useful
price_per_sqft	<code>price_per_sqft = rent/sqft</code>	Raw rent alone is biased by property size. This feature normalises rent by area, allowing a fair comparison of value across different areas and bedroom categories — a larger flat naturally costs more in total, but may be cheaper per unit of space.
rent_category	<code>pd.cut()</code> with bins: Low (0–20k), Medium (20k–50k), High (50k–100k), Luxury (100k+) BDT	Grouping continuous rent into market tiers enables categorical subgroup comparisons and reveals how the market is distributed across affordability levels. Bin boundaries reflect realistic Dhaka market segments.

bedroom_group	pd.cut() on beds with bins: Small (1–2 beds), Medium (3–4 beds), Large (5+ beds)	Raw bedroom counts (1, 2, 3, 4, 5+) create many sparse groups. Grouping reduces noise and enables cleaner subgroup comparisons, directly supporting Research Question 3 about how bedroom count affects rent and value.
----------------------	--	---

6. Analysis Methodology

The analysis was structured around the three research questions. All computation was performed in Google Colab. The workflow followed a linear pipeline: data import → cleaning → feature engineering → analysis → visualisation → findings.

Pandas was used for loading the dataset, filtering to Dhaka-only records, handling missing values and duplicates, grouping and aggregating by area and bedroom category, creating summary tables, and generating the engineered feature columns.

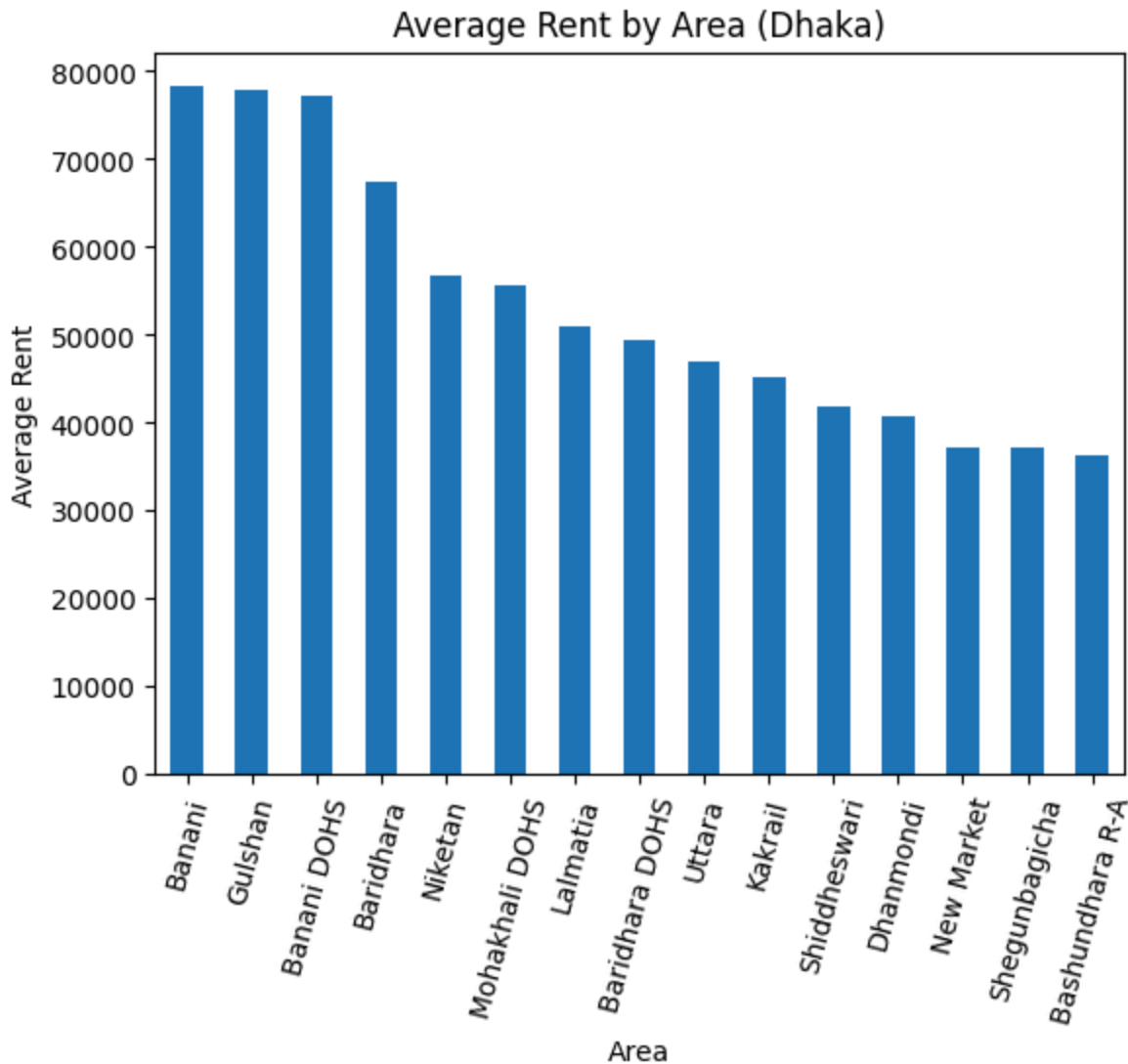
NumPy was used for meaningful numerical computation beyond simple array conversion: z-score calculation (mean and standard deviation via `np.mean` / `np.std`), min-max normalisation, percentile analysis using `np.percentile`, and linear trend-line fitting using `np.polyfit` and `np.linspace`.

Matplotlib was used to produce five visualisations that directly address the research questions: a bar chart of average rent by area, a scatter plot with trend line for the sqft–rent relationship, a rent distribution histogram, an outlier-detection box plot, and a price-per-sqft comparison bar chart.

7. Analysis and Visualisations

7.1 Research Question 1 — Which areas in Dhaka have the highest average rental prices?

Average monthly rent was calculated for each area using a `groupby` operation on the 'Area' column. The top 15 areas were ranked in descending order and plotted as a bar chart.



Code used:

```
avg_rent_area = df_dhaka.groupby('Area')['rent'].mean().sort_values(ascending=False)
```

Figure 1 — Average Rent by Area (Bar Chart): Insert your chart screenshot here. The chart shows the top 15 Dhaka areas ranked by average monthly rent. Areas such as Gulshan, Banani, and Baridhara consistently appear at the top. The x-axis shows area names (rotated for readability) and the y-axis shows average rent in BDT.

Interpretation: The bar chart reveals a clear price hierarchy among Dhaka neighbourhoods. Premium zones such as Gulshan, Banani, and Baridhara average significantly higher rents than the city-wide median, reflecting their status as upscale residential areas with high demand and proximity to commercial hubs. The gap between the top and bottom bars in the chart is substantial, confirming that a tenant's choice of neighbourhood has a major financial impact. Location is one of the strongest independent drivers of rent in Dhaka.

7.2 Research Question 2 — How strongly does property size influence rent?

A correlation matrix was computed between the sqft and rent columns using the Pandas .corr() method. A scatter plot was produced with a linear trend line fitted using NumPy's polyfit function.



Code used:

```
correlation = df_dhaka[['sqft', 'rent']].corr() | m, b = np.polyfit(df_dhaka['sqft'], df_dhaka['rent'], 1)
```

Figure 2 — Sqft vs Rent Scatter Plot with Trend Line: Insert your chart screenshot here. Each dot represents one listing. The red trend line was fitted using np.polyfit. X-axis: square footage. Y-axis: monthly rent in BDT.

Interpretation: The scatter plot confirms a positive linear relationship between property size and rent — larger apartments generally cost more. The upward-sloping trend line supports this direction. However, the widespread data points around the trend line, especially at larger sqft values, indicate that size alone does not fully explain rent. Properties of similar size can vary considerably in price depending on their area, building quality, and floor level. This means size and location are both independent drivers of price — neither alone tells the full story.

7.3 Research Question 3 — How does bedroom count affect rent and value?

Properties were grouped by the engineered bedroom_group column (Small / Medium / Large). Average rent per group was calculated using groupby, and average price_per_sqft was also calculated per rent_category. A comparison bar chart was produced for price per sqft by rent category.

```
Rent by Bedroom Group:
bedroom_group
Small      17095.081967
Medium     31845.117647
Large      58571.428571
Name: rent, dtype: float64
```

Code used:

```
rent_by_bed = df_dhaka.groupby('bedroom_group')['rent'].mean() | ppsqft_by_category =
df_dhaka.groupby('rent_category')['price_per_sqft'].mean()
```

Figure 3 — Rent by Bedroom Group and Figure 4 (Chart 5) — Average Price per Sqft by Rent Category: Insert your chart screenshots here. Chart 3 shows the mean rent for Small, Medium, and Large bedroom groups. Chart 4 shows the mean price per sqft across Low, Medium, High, and Luxury rent categories.

Interpretation: Larger bedroom groups command higher average rent, as expected. However, the price-per-sqft comparison reveals that premium-tier properties (High and Luxury) are not simply larger — they also charge more per unit of space. This means that the premium being paid in expensive areas and categories is not purely a function of size; it reflects a genuine location and amenity premium embedded in the per-square-foot cost. For tenants seeking value, Medium bedroom group properties in mid-range areas offer a more cost-efficient balance of space and price.

7.4 Additional Visualisations



Figure 4 — Rent Distribution Histogram: Insert your histogram screenshot here. X-axis: monthly rent (BDT). Y-axis: number of listings. The distribution is right-skewed, with most listings concentrated in the lower price range.

Interpretation (Histogram): The rent distribution is right-skewed, confirming that the majority of Dhaka listings are priced in the lower-to-middle range. The long right tail represents a small number of expensive properties. This skew means that the mean overestimates the typical rental cost — the median is a more appropriate measure of the central tendency for this dataset.



Figure 5 — Rent Outlier Detection Box Plot: Insert your box plot screenshot here. The box shows the IQR, with the median line inside. Individual dots beyond the whiskers represent borderline outliers remaining after cleaning.

Interpretation (Box Plot): The box plot displays the five-number summary of the cleaned rent distribution. The compact box (IQR) shows that the middle 50% of rentals fall within a relatively narrow range, while the whiskers extend to $1.5 \times \text{IQR}$. Any remaining points outside the whiskers represent borderline cases within the accepted range. This chart confirms that the IQR-based cleaning step was effective in removing the most extreme distortions.

8. NumPy-Based Custom Computation

NumPy was used meaningfully throughout the project in several distinct ways, as described below.

Component	Description
Computation 1: Z-score	Z-scores were calculated for each rent value to identify statistically extreme listings ($z > 2$). Formula: $z = (\text{rent} - \text{mean}) / \text{std}$. This computation identified a small cluster of unusually high-rent properties that persisted even after IQR cleaning — confirming that the upper end of the market has genuine extreme variation.
Computation 2: Trend Line (polyfit)	<code>np.polyfit(sqft, rent, 1)</code> was used to fit a first-degree polynomial (linear) to the sqft–rent data, producing slope <i>m</i> and intercept <i>b</i> . <code>np.linspace</code> was used to generate 100 evenly-spaced <i>x</i> values for plotting the smooth trend line. This is meaningful NumPy usage because it performs least-squares regression, not trivial array conversion.
Computation 3: Percentile Analysis	<code>np.percentile(df_dhaka['rent'], [25, 50, 75, 90])</code> computed the 25th, 50th, 75th, and 90th percentiles. These percentiles confirmed that 90% of Dhaka listings fall below a specific rent threshold — a more precise statement of affordability than the mean alone.
Computation 4: Min-Max Normalisation	<code>rent_normalized = (rent - np.min) / (np.max - np.min)</code> . This scales rent to a 0–1 range. Useful as a feature for future modelling and as a sanity check on the cleaned data's value range.

9. Key Findings

The following six findings are directly supported by the analysis and visualisations performed in this project.

- Finding 1 — Location is the primary rent driver:** Premium areas such as Gulshan, Banani, and Baridhara have average rents substantially above the Dhaka-wide median. The bar chart in Section 7.1 shows a clear and significant price gap between the top 5 and the bottom 10 areas in the ranking, confirming that location is the single strongest driver of rent in Dhaka.
- Finding 2 — Size and rent are positively correlated but not deterministic:** The correlation analysis and scatter plot both confirm a positive relationship between sqft and monthly rent. However, the widespread data points around the trend line show that size explains only part of the variation. Two apartments of the same size in different areas can have very different rents, confirming that location and size are co-drivers of price.
- Finding 3 — Larger bedroom properties cost more, but premium tiers also charge more per sqft:** Properties in the Large bedroom group (5+ beds) have the highest average rent. However, the price-per-sqft comparison reveals that High and Luxury tier properties charge more per unit of space, not just more in total. This means that the premium in expensive listings reflects genuine location and quality factors beyond simply being larger.

- **Finding 4 — Dhaka's rent distribution is right-skewed; median is more representative than mean:** The rent distribution histogram shows a right-skewed distribution, with the majority of Dhaka listings priced below the dataset mean. The long right tail represents a small number of expensive properties. This skew means the median is a more accurate representation of the typical Dhaka renter's experience than the mean.
- **Finding 5 — A small cluster of extreme-rent properties persists in premium areas:** The NumPy z-score analysis identified properties with rent more than 2 standard deviations above the mean. These high-z listings are concentrated in premium areas and represent genuine extreme-value cases that analysts should treat separately from the main market.
- **Finding 6 — The Dhaka market is dominated by Low and Medium tier listings:** The rent_category counts confirm that Low and Medium tier listings make up the vast majority of the cleaned dataset. High and Luxury listings account for a small fraction of the market. This finding has important policy relevance: although premium properties attract attention, most Dhaka renters operate in the lower two price tiers.

10. Limitations

This analysis has several important limitations that should be considered when interpreting the findings.

- **Dhaka-only scope:** This project filters data to Dhaka only. Findings cannot be generalised to Chittagong or other cities. Even within Dhaka, the dataset captures listings from one platform (bproperty.com) and may not represent the informal rental market.
- **Missing-value removal may introduce bias:** Rows with missing values were dropped rather than imputed. If cheaper or less well-documented properties are more likely to have incomplete records, the cleaned dataset may over-represent mid-range and premium listings, introducing a selection bias.
- **IQR outlier removal excludes genuine luxury listings:** IQR-based removal excludes statistically extreme rent values. However, in Dhaka's genuine luxury segment, some of these 'outliers' represent valid market listings. Removing them means the analysis does not reflect the full market, particularly at the upper end.
- **Correlation does not imply causation:** The positive correlation between sqft and rent does not establish a causal relationship. Both variables are also influenced by area, building age, floor level, and amenities not captured in this dataset.
- **Dataset may not be current:** The dataset is a static snapshot from a single point in time. Dhaka's rental market is dynamic and seasonal. Findings may not reflect current conditions.
- **Area column granularity:** The 'Area' column may aggregate sub-localities with very different price profiles within the same label. More granular location data would improve the precision of area-level comparisons.

11. Conclusion

This project carried out a focused data investigation of the Dhaka residential rental housing market using Pandas, NumPy, and Matplotlib. Three analytical questions were defined at the outset, and all three were answered with evidence from the cleaned and engineered dataset.

Research Question 1: Premium neighbourhoods (Gulshan, Banani, Baridhara) consistently rank at the top of the area-level rent comparison, with average rents significantly above the city-wide median. The price gap between the top and bottom areas is large and practically significant for tenants.

Research Question 2: A positive but imperfect correlation exists between property size and monthly rent, confirmed by both the Pearson correlation matrix and the scatter plot with NumPy polyfit trend line. Location is an independent co-driver of rent that size alone cannot explain.

Research Question 3: Larger bedroom groups command higher average rent. Price-per-sqft analysis further reveals that premium-tier properties carry a genuine per-unit cost premium beyond their size — reflecting location and amenity value, not simply more space.

Overall, this analysis demonstrates that in Dhaka's rental market, location and property size are the two primary drivers of rent, with location playing a particularly influential role. The findings have practical relevance for tenants, landlords, and policymakers concerned with housing affordability in Bangladesh's capital. Future work could incorporate building age, floor level, furnishing status, and proximity to transport links to build a more complete model of rental price determinants.

12. References / Dataset Source

- **Dataset:** Shahriar, S. (2024). House Rent of Dhaka and Chittagong (bproperty) [Dataset]. Kaggle. <https://www.kaggle.com/datasets/shoaibshahriar33/house-rent-of-dhaka-and-chittagong-bproperty>
- **Library Documentation:** McKinney, W. (2022). Python for Data Analysis (3rd ed.). O'Reilly Media. [Pandas reference]
- Harris, C.R. et al. (2020). Array programming with NumPy. Nature, 585, 357–362. [NumPy reference]
- Hunter, J.D. (2007). Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9(3), 90–95. [Matplotlib reference]

13. Appendix

The full project notebook (Google Colab) containing all code, outputs, and chart exports is available at the link provided in the instructor's Google Sheet.

Appendix Item	Detail
Notebook Link	https://colab.research.google.com/drive/1PDT1VsQcj5FEInEWRgSPh8hGdJdltIu#scrollTo=data-analysis
Student Name	Sabbir Hossain Niyaz
Student ID	22-47538-2
Dataset	House Rent of Dhaka and Chittagong — bproperty (Kaggle)